Computer Programming II (CSE2035) Week 6 Lab Session: Object Oriented Programming Part 2

Dept. of Computer Science and Engineering
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Exercise 0: Lecture Class

- **Let's write a class that manages lecture information**
- **Properties, constructor, and `GetCode()` are given in `lecture.hpp` file (do not fix this file)**

```
enum Grade { A, B, C, NA };
...
class Lecture {
private:
    char code[MAX_CODE_LEN + 1];
    int student_cnt;
    int ids[MAX_STUDENT];
    Grade grades[MAX_STUDENT];
public:
    Lecture(const char * _code): student_cnt(0) {
        strncpy(code, _code, MAX_CODE_LEN);
        code[MAX_CODE_LEN] = '\0';
    }
```

Exercise 0: Lecture Class

- **Implement the following functions in `lecture.cpp`**

```
// Add a student whose ID is 'id'. The student's grade must be
// initialized as 'NA'. Store the information of the student at
// 'ids[student_cnt]' and 'grades[student_cnt]'. After adding the
// student, increase 'student_cnt' by one and return true. If the
// lecture is already full or 'id' is already present in this
// lecture, don't add the student and just return false.
bool AddStudent(int id);

// Update the grade of the student whose ID is 'id' into 'grade'. If
// such a student doesn't exist, just return false.
bool SetGradeOf(int id, Grade grade);

// Get the grade of the student whose ID is 'id', and store it at
// '*pGrade'. If such a student doesn't exist, just return false.
bool GetGradeOf(int id, Grade *pGrade);
```

Exercise 0: Lecture Class

- **In `main.cpp`, you can find a test function for `Lecture`**
  - You can freely change this function to test your class implementation in various ways

```
void test() {
    Lecture lec1("CSE2015");
    Lecture lec2("CSE2035");
    Grade grade;

    // Test with the first lecture.
    lec1.AddStudent(20220001);
    lec1.AddStudent(20220002);
    lec1.SetGradeOf(20220002, A);

    // This should work and print "A" as grade.
    if (lec1.GetGradeOf(20220002, &grade)) {
        cout << "Grade of 20220002: " << GradeToString(grade) << endl;
    }
    ...
```

Lab Assignment Submission Instructions

- **Create “WeekX_Lab_yourID_name” folder**
  - E.g., Week6_Lab_20221234_문의현
- **Put your completed works (e.g., codes/files/programs) to the folder**
- **Compress (Zip) your folder**
  - E.g., Week6_Lab_20221234_문의현.zip
- **Submit/Upload your Zip file to the Cyber Campus in the “Assignments” section**
- **Due dates: All the lab assignments are due by Wednesday at 11:59pm, right after prior week’s lab session on Thursday**
  - Late submissions are not accepted